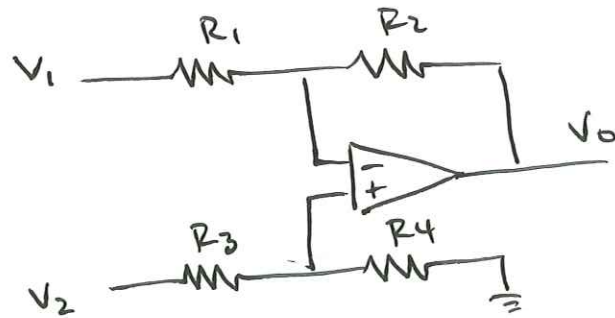


Difference Amplifier



$$\rightarrow V_0 = \frac{R_4}{R_1} \left(\frac{R_1 + R_2}{R_3 + R_4} \right) V_2 - \frac{R_2}{R_1} V_1$$

Typically desired case is:

$$V_0 = A (V_2 - V_1)$$

$$A = \frac{R_2}{R_1}$$

$$R_3 = R_1 \\ R_4 = R_2$$

what if $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ is desired BUT

in actuality: $\frac{R_1}{R_2} = \frac{R_1 + \delta_3}{R_2 + \delta_4} = \frac{R_3' = R_1'}{R_4' = R_2'}$ what happens?

$$\text{let } \frac{R_4}{R_3} = \frac{R_2'}{R_1'} = \frac{R_2}{R_1} (1 - \epsilon)$$

$\epsilon \rightarrow$ some small error value

from ideal equation ...

$$V_0 = \underbrace{\frac{R_2}{R_1} \left[1 - \frac{R_1 + 2R_2}{R_1 + R_2} \frac{\epsilon}{2} \right]}_{\text{Diff. mode gain } A_{dm}} (V_2 - V_1) + \underbrace{\frac{R_2}{R_1 + R_2} \epsilon}_{\text{common mode gain, } A_{cm}} \underbrace{\frac{V_2 + V_1}{2}}_{\text{Average. } V_{cm} = \text{"common mode" voltage.}}$$

$$V_0 = A_{dm} V_{dm} + A_{cm} V_{cm}$$

$$V_1 = V_{cm} - V_{dm}/2$$

$$V_2 = V_{cm} + V_{dm}/2$$

$$V_{dm} = V_2 - V_1$$

$$V_{cm} = \frac{V_2 + V_1}{2}$$

Common Mode Rejection Ratio, CMRR [dB]

$$\text{CMRR}_{\text{dB}} = 20 \log_{10} \left(\frac{A_{\text{DM}}}{A_{\text{CM}}} \right)$$

ideally $A_{\text{CM}} \rightarrow 0$, $\text{CMRR}_{\text{dB}} \rightarrow \infty$

Reasonable targets usually $\geq 100\text{dB}$

for a Diff amp with Resistor Ratio error ε

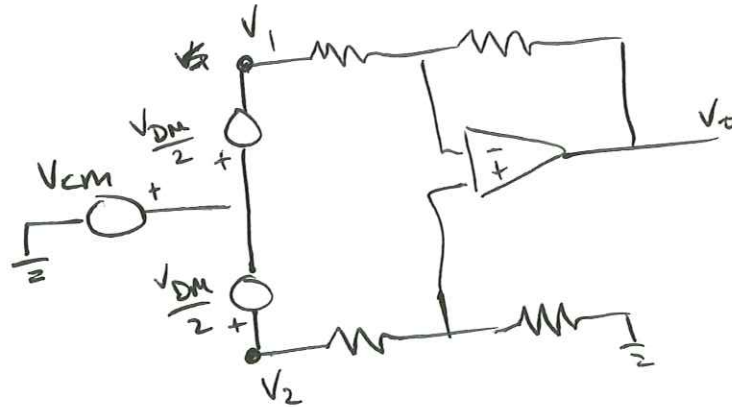
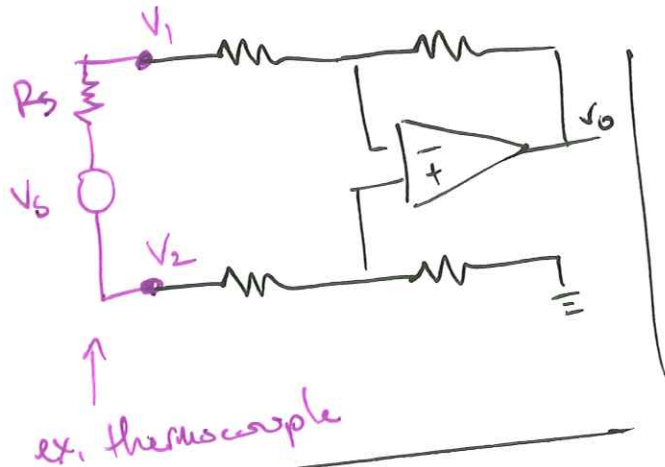
$$\text{CMRR}_{\text{dB}} \approx 20 \log_{10} \left(\frac{R_2/R_1}{\varepsilon R_2/R_1 + R_2} \right) = 20 \log_{10} \left(\left(1 + R_2/R_1 \right) / \varepsilon \right)$$

$$= 20 \log_{10} \left((1 + A_{\text{ideal}}) / \varepsilon \right)$$

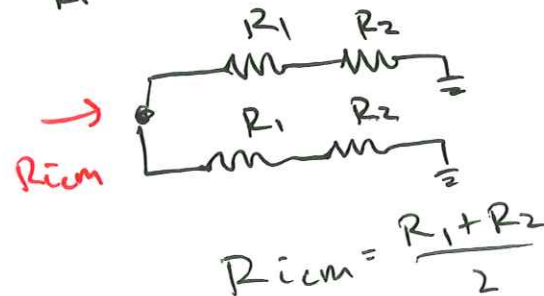
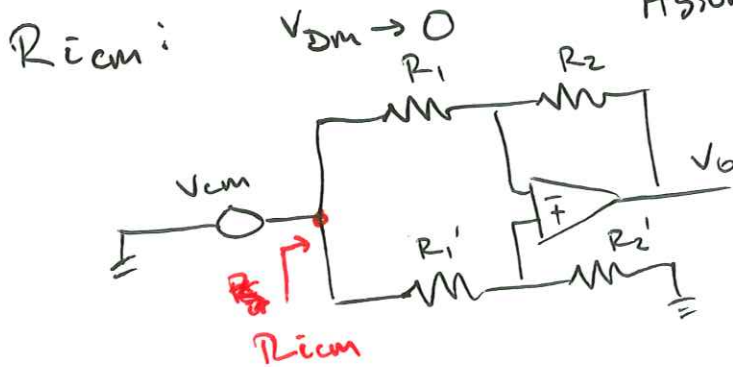
$\uparrow A_{\text{DM ideal}}$

In design, expect max ε from given tolerances

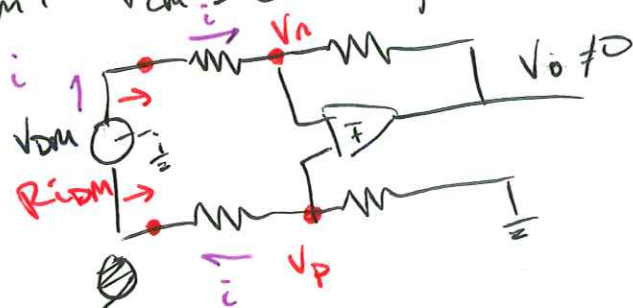
Difference Amp Resistances



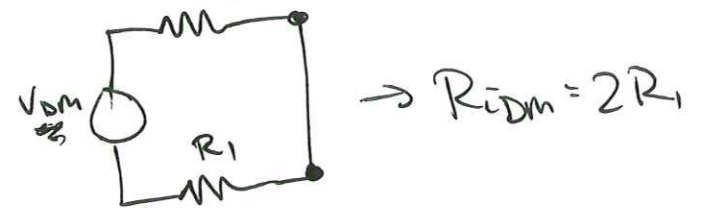
Assume $\frac{R_2}{R_1} = \frac{R_2'}{R_1'} \rightarrow$ if $V_{DM} = 0$, then $V_o = 0$



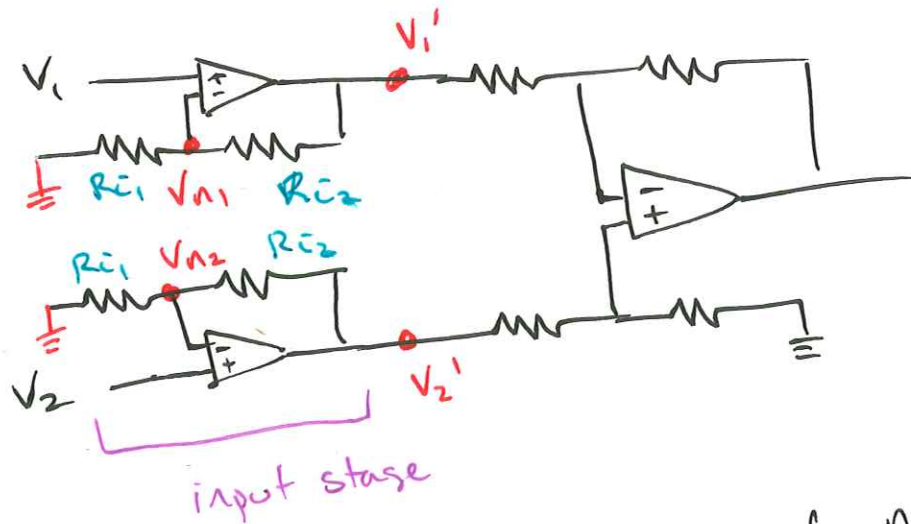
R_{idm} : $V_{cm} \rightarrow 0$ Again considering ideal $\rightarrow V_o = \frac{R_2}{R_1} V_{DM}$



$V_p = V_n$
effectively



Isolate Sensors/Inputs from R_{icm}/R_{idm}



if input stage has a gain of A_1 :

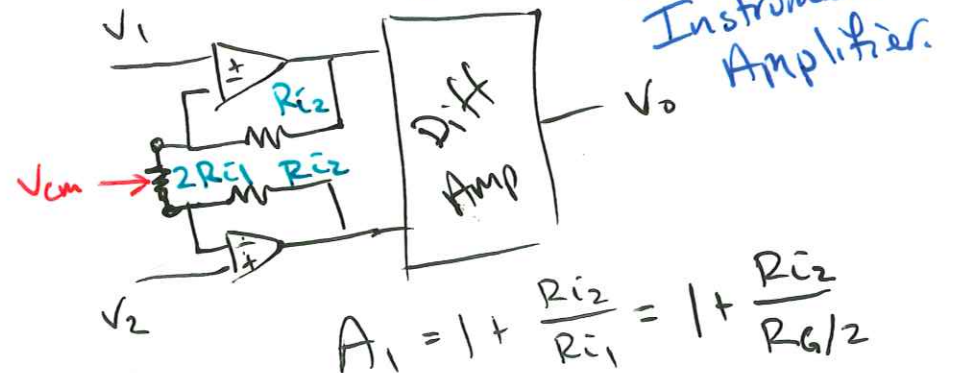
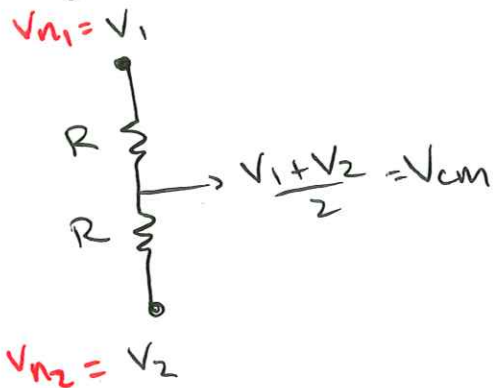
$$V_1' = A_1 V_1 = A_1 V_{cm} - A_1 V_{om}/2$$

$$V_2' = A_1 V_2 = A_1 V_{cm} + A_1 V_{om}/2$$

Don't want to do this!
likely saturate amps.

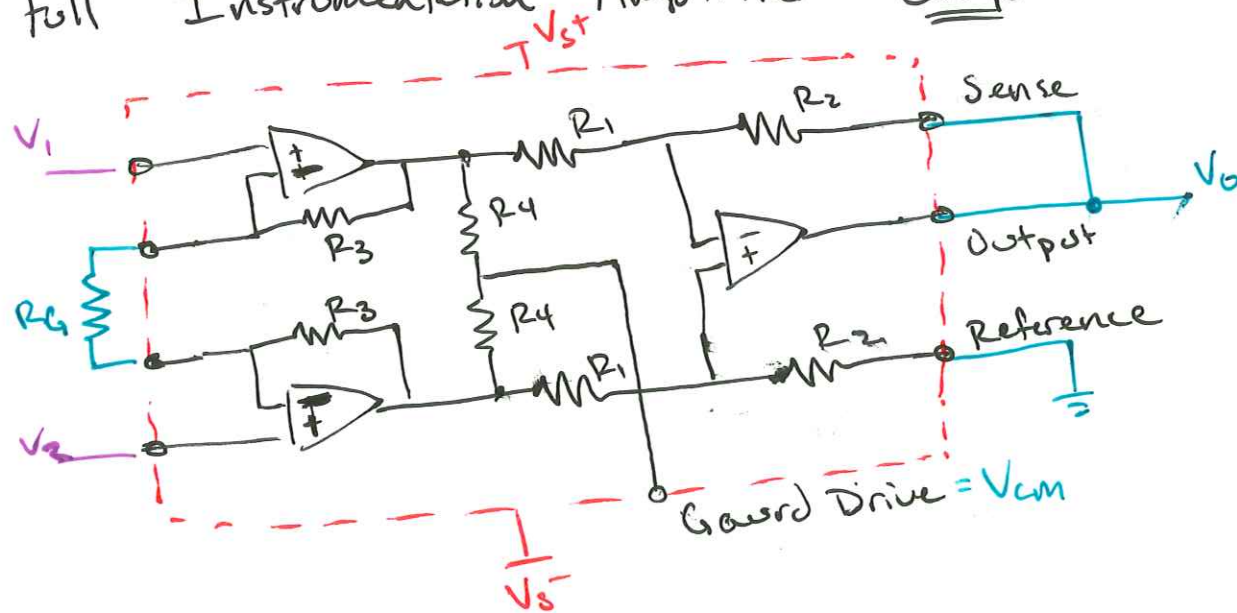
To fix: Use V_{cm} as input stage reference voltage instead of ground.

Assuming virtual short: $V_{n1} = V_1$, $V_{n2} = V_2$



$$A_1 = 1 + \frac{R_{i2}}{R_{i1}} = 1 + \frac{R_{i2}}{R_G/2}$$

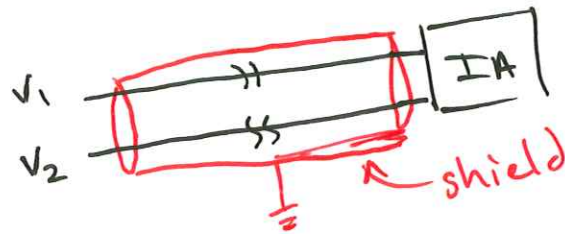
full Instrumentation Amplifier Chip



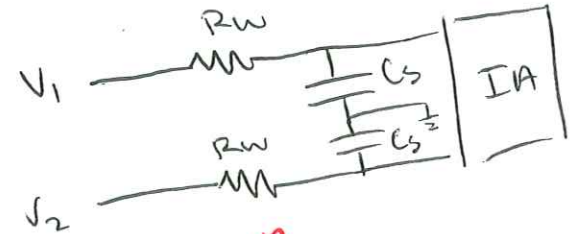
ideal and usually effective.

$$A_{\text{den}} = \frac{R_2}{R_1} \left(1 + \frac{2R_3}{R_G} \right)$$

Use of Guard Drive ...
consider inputs with long wires



→ lumped model of inputs:



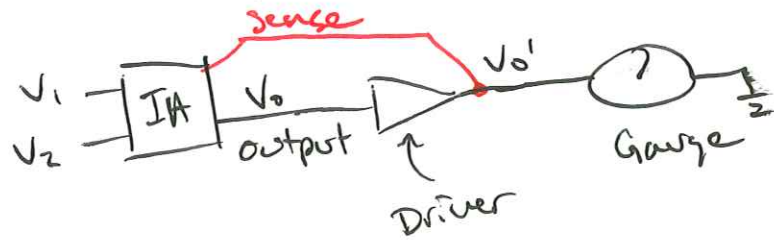
↑
Shield causes low pass filtering to occur!

Avoid filtering: → reduce/eliminate voltage difference between wires & shield

→ Drive shield with Guard Drive!

Sense Terminal

Compensate for additional circuitry on output term.

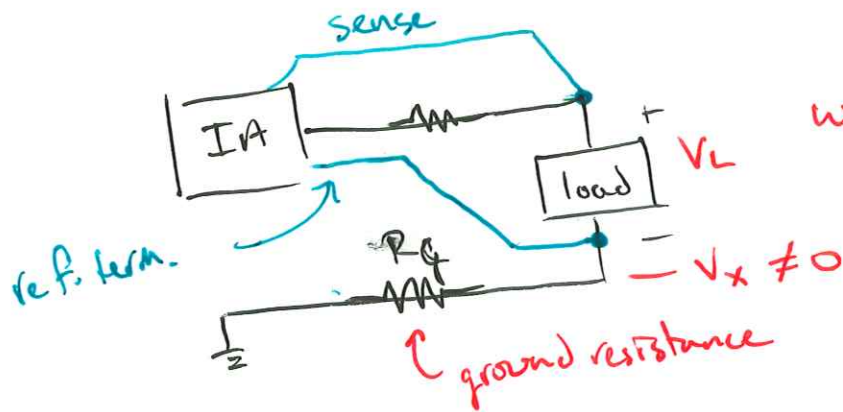


want V_0' to be the correct value: $V_0' = A_{om} V_{om}$

Sense terminal will eliminate induced Driver errors

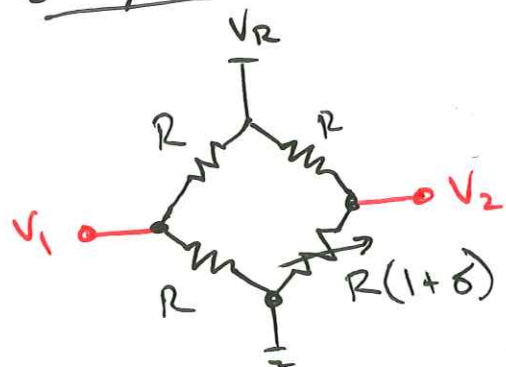
Reference Terminal

$$\rightarrow V_0 = \frac{R_2}{R_1} \left(1 + \frac{2R_3}{R_4} \right) V_{DM} + V_{REF} \leftarrow \text{voltage applied to this pin}$$



$$\text{want } V_L = \frac{R_2}{R_1} \left(1 + \frac{2R_3}{R_4} \right) V_{om}$$

Bridges



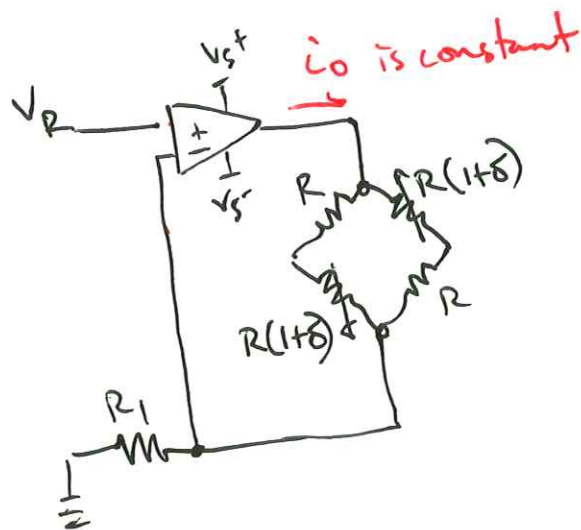
$$V_1 = \frac{1}{2} V_R$$

$$V_2 = \frac{1+\delta}{2+\delta} V_R$$

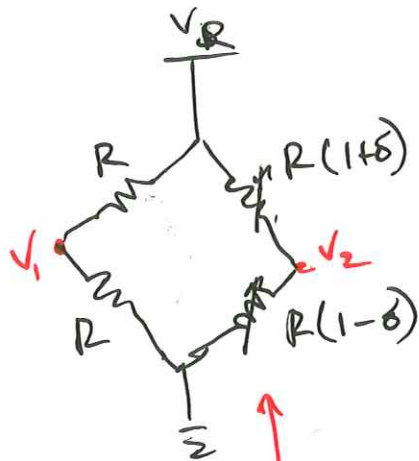
$$V_2 - V_1 = \frac{\delta V_R}{2(2+\delta)}$$

$$\text{if } \delta \ll 2$$

$$\text{then } V_2 - V_1 \approx \frac{\delta V_R}{4}$$



$$V_2 - V_1 = \frac{\delta R}{2R_1} V_R$$



$$\rightarrow V_2 - V_1 = -\frac{\delta}{2} V_R$$

↑ strain gauges in bending.

